

# The Goods

[ATITD Wiki](#) • [Manifesto](#) • [Pricing Explained](#) • [Inventory/Valuation](#) • [Trading Procedures](#) • [Tellers](#) • [Admin](#)

Login:

## How we value goods

The value we assign to a commodity depends on one thing only: the amount of that commodity we have in stock. The basic unit of our pricing is the **First Good**, which is the value we assign to the first unit of any commodity we add to our stores. It doesn't matter what the good is. As far as we're concerned, the value of a diamond is the same as the value of a carrot, if we have only one diamond and one carrot in stock. (We expect to have more than one carrot in stock most of the time.)

The value we assign to having  $n$  units of a commodity in our stores is  $\log_2(n+1)$  First Goods. Suppose we have  $j$  units of a commodity. If you want to sell us  $k$  more, we will value those  $k$  items at a total of  $\log_2(j+k+1) - \log_2(j+1)$  First Goods; if you wish to buy  $k$  of the  $j$  items we have in stock, we will ask for  $\log_2(j+1) - \log_2(j-k+1)$  First Goods in return. The [Valuation page](#) has a calculator you can use to value goods at the current prices.

To explain in simple terms what this means:

- The more we have of a commodity, the less value we assign to it.
- The first unit of a commodity has the value of 1 First Good. The next two units together have a total value of 1 First Good. So do the four after that, the eight after that, and so on.
- We will always pay exactly 1 First Good for one more than our entire stock of an item. (For example, if we have 437 coal, we will pay 1 First Good for 438 coal.) Likewise, if you want to buy more than half of our stock of an item, it will cost at least 1 First Good.
- The value of a unit does not "plateau". If you buy 100 glass rods, you will not be simply paying 100 times a base price; the price per unit will always vary with the quantity. We will try to provide tools to make the price calculation easy.
- However, constant proportions of our stock will tend to converge to a constant value. For example, 10% of our entire stock of an item will be worth about 0.152 First Goods, regardless of how much we have in stock.
- Consequently, relative prices will tend to remain stable as absolute quantities vary. If we have  $n$  times as many of item A in stock as item B, then the price of item B will be approximately  $n$  times the price of item A. The more units we have in stock, the closer the approximation.
- There is no limit to the quantities we will buy. With the formula given, it will in theory always be possible to buy anything we have in stock with whatever commodity you can provide, as long as you can provide it in sufficiently large quantities. However, the quantities required can quickly reach levels where this is impracticable. (See also the [Trading Procedures](#) page on storage fees for large trades.)
- Prices are self-correcting. If we value an item too low, we will tend to sell more of it than we do other goods, increasing its relative price. If we value it too high, we will tend to buy more of it, lowering its relative price.

[Examples of how the pricing system works](#)

## Ripe for abuse?

It may seem at first that the rules listed above provide ways for people to easily rip us off. After all, if you can pay for a large sapphire with deben of dirt, doesn't that indicate something is wrong with the system? Because all First Goods are valued equally, the first deben of dirt we receive will be valued at the same price as the first Large Sapphire we receive. The first person who sells us dirt will be able to take Goodscrip in payment and keep it until someone sells us a Large Sapphire.

We do not regard this as cheating the system. We regard it as the rewards of long-term investment. In general, we expect to acquire lower-value items well before higher-value ones. Before anyone sells us a Large Sapphire, we'll have to have lower-valued goods in sufficient quantity to make it worthwhile. Thus, in order to make the trade described above, someone would have to keep their Goodscrip for a long time, and therefore leave 1 First Good worth of goods in our stores for a long time. This is exactly the sort of behavior we wish to encourage.

We do expect this system to result in large stockpiles of dirt, dung, and other low-value commodities. However, we also expect it to rapidly become impractical for people to sell us more of those things.